



FOR YOUR DEVELOPER

Every piece of technology the company needs

This document is for your developer.

Status labels apply throughout: KNOWN, VERIFIED, ASSUMPTION, ESTIMATE, TO VALIDATE, PROPOSED. Nothing in this document claims traction, approvals, partners or revenue that do not exist. Governing file: FACTS_BASE.

PREPARED FOR

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This document is for your developer.

One table for the whole company: what each thing is for, what to use, roughly what it costs a month, and what else would do.

In one minute

- Vytalix has two separate estates: cheap bought tools for the business, and one built platform for MaternaLink.
- Buy everything that is not the product. Build only the product.
- The business tools cost £90–£250 a month at the start, rising to £460–£1,560 once the build is running.
- MaternaLink runs on AWS in London for £350–£700 a month at first release.
- The "do not build yet" list at the end matters as much as the buy list. Keep it on the agenda.

What to do

Action	Who	By when
On incorporation: domain, Google Workspace, email security, password manager, HubSpot free, Xero, Webflow, Plausible, ICO registration	Founder	One week of setup, under £300 a month
Set up the AWS organisation, infrastructure-as-code baseline and GitHub before the build starts	Developer	Before build week 1
Write down the AI policy and have the founder sign it	Founder	Before any tool is used on client work
Choose the e-learning platform when the first course is scheduled, not before	Founder	Month 6
Book Cyber Essentials for the quarter in which the MaternaLink test environment exists	Developer	2027

How it fits together

flowchart TB

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subgraph Bought[Bought: business tools, no patient data]
  WEB[Website: Webflow now, Next.js later] --> CRM[HubSpot]
  LMS[Courses: Thinkific or LearnWorlds] --> FIN[Xero, Stripe, GoCardless]
  GW[Google Workspace: mail, docs, sign-in] --> CRM
  GW --> SEC[1Password, device management]
  WEB --> AN[Plausible analytics]
  FIN --> NGP[Paystack or Flutterwave, Nigeria]
  DOCS[PandaDoc contracts]
end

subgraph Built[Built: MaternaLink, AWS London, all patient data]
  PWA[Woman and staff web apps] --> API[API and workers]
  API --> PG[(PostgreSQL)]
  API --> S3[(File storage)]
  API --> KC[Keycloak sign-in]
  API --> CH[SMS and WhatsApp adapters]
  PG --> MB[Metabase dashboards]
end

subgraph Later[Deferred until someone asks for it]
  PORTAL[Customer portal]
  FHIR[NHS interfaces]
  DHIS[DHIS2 export]
  NGH[Nigeria hosting]
  DW[Data warehouse]
end

API --> FHIR
API --> DHIS
API --> NGH
PG --> DW
KC --> PORTAL

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Caption: business tools on the left are bought and hold no patient data; the product platform on the right is built, and patient data never leaves it.

The whole company's technology in one table

All prices are ESTIMATE (our best guess, not a fact) in pounds at September 2026, from public list prices. Check every one before buying; vendors rename plans often.

What it is for	What to use	Roughly per month	The alternative
Company website	Webflow with its CMS plan in year 1. Next.js with a headless CMS once engineers exist in 2027	£18–£25, plus £10–£15 a year for the domain	Framer, similar and design-led. WordPress, cheap but a security burden. Squarespace, too limited
Marketing content	Webflow's own CMS now, then Sanity or Payload later. MaternaLink's clinical content is not here: it lives in the product with its own two-person approval	Included, then £0–£80	Contentful, pricier. Strapi, self-hosted
Sales and contacts	HubSpot, free tier then Starter. The investor list imports as its own pipeline	£0, then £15–£20 per seat	Pipedrive £12–£14 per seat, sales only. Attio £25+. Salesforce, no

What it is for	What to use	Roughly per month	The alternative
Customer portal	Do not build in year 1. Serve consulting clients with shared drives and PandaDoc, learners inside the course platform, MaternaLink customers inside the product's admin screens	£0 now, £30k–£60k to build later	HubSpot's own portal for tickets
Selling courses	Thinkific if the first courses are for individuals, LearnWorlds if a corporate or NHS group licence is signed first	£30–£120 Thinkific, £25–£250 LearnWorlds	Teachable, weaker on corporate. Moodle, only when a university or NHS buyer demands it: £80–£150 hosted, or £30–£100 self-hosted plus admin time. Custom build £40k–£80k, not before year 3
The MaternaLink product	TypeScript throughout; Next.js web apps; a NestJS API; PostgreSQL with row-level security; Redis for jobs; S3; Keycloak; AWS eu-west-2 (London); Terraform; GitHub Actions; SMS adapters for Africa's Talking, Termii and Twilio	£350–£700 at first release; £1,000–£2,500 at 5 organisations and 20,000 women	Azure UK South, equivalent and worth choosing if the first NHS customer is Microsoft-centred. Do not run both. GCP London has fewer NHS references
Product admin and dashboards	Admin lives inside the MaternaLink staff app. Business dashboards on Metabase, self-hosted, reading a copy of the database. Company numbers start as a spreadsheet fed by HubSpot, Xero and course exports	£30–£50 for Metabase, £0 for the spreadsheet	Retool, fast but data leaves the region unless self-hosted. Power BI if the buyers are Microsoft-heavy
Analytics	Plausible for the website, hosted in the EU with no cookies. For the product, an events table inside our own database holding no personal data, read by Metabase	£9–£15, product side included in cloud cost	Fathom. Google Analytics is free but brings a consent banner and data-transfer questions. Amplitude and Mixpanel are excluded: data leaves the region
Staff sign-in	Google Workspace as the identity provider for every business tool, with security keys for admins. A company domain replaces the personal Gmail on day one	£5–£11 per user	Microsoft 365 is a legitimate choice because NHS buyers are Microsoft-heavy. Decide once
MaternaLink sign-in	Keycloak, self-hosted in London, for staff. Phone code plus device PIN for women	£40–£70	Auth0 and Clerk are easier but their UK data residency is unconfirmed. Cognito is cheap but clumsy to federate
Taking payments	Stripe for cards and subscriptions, usually through the course platform. Xero invoicing with GoCardless direct debit for retainers. Paystack or Flutterwave for Nigeria, which needs a Nigerian entity or a partner	Xero £15–£40. Fees: about 1.5% plus 20p on UK cards; GoCardless about 1% capped; Paystack about 1.5% plus ₦100 over ₦2,500	Paddle handles global VAT itself at a higher percentage. Interswitch or Monnify in Nigeria. Government and donor contracts are invoice and bank transfer only
Joining other systems	Our own documented API first. FHIR UK Core read-only view in version 2. DHIS2 export for Nigeria. Hospital record interfaces where the vendor supports them	Mostly effort: £20k–£40k per NHS integration, £30k–£50k for the FHIR view	Zapier or Make for business tools only, never for patient data
Cloud and databases	AWS eu-west-2 with separate accounts for production, non-production, backups and security. PostgreSQL everywhere. Redis for queues only. S3 with write-once storage for audit copies	See the product row. A Nigeria cell adds £350–£1,500	Azure UK South. Hetzner or OVH are cheap but weak for NHS assurance. Vercel and Supabase are fine for the corporate site, not for patient data

What it is for	What to use	Roughly per month	The alternative
AI tools	Business tiers with no-training terms, for drafting marketing and course content, code help, internal meeting notes and translation first drafts	£20–£60 per user, plus £10–£40 for coding assistants	None needed. What matters is the policy, not the vendor
Security	Google Workspace sign-in with passkeys. Device management. 1Password. Email authentication set to reject forgeries. Cyber Essentials, then Plus, then the NHS toolkit	1Password £6–£8 per user; devices £0–£8 each; certification £2k–£4k a year	Bitwarden instead of 1Password. ISO 27001 deferred to year 2 or 3 at £15k–£40k
Data governance and backups	ICO registration on incorporation, a fractional data protection officer, records of processing, retention schedule, DPIA templates and a sub-processor register. Product backups: point-in-time recovery, 35-day snapshots, a copy in a separate UK account, restore drills every quarter. Business tools export monthly to encrypted storage; code sits in GitHub with branch protection and a nightly mirror	DPO £500–£1,500; ICO fee £52–£78 a year; SaaS backup £3–£6 per user; product backup included in cloud	None. Neither is optional once real data exists

Annual items on top (ESTIMATE): Cyber Essentials £300–£600; Cyber Essentials Plus £1,500–£3,000; penetration test £6,000–£12,000; accessibility audit £3,000–£6,000; ICO fee £52–£78; insurance £1,000–£3,000.

The lean first-year setup

Phase A is months 1 to 3, services only, two paid seats. Phase B is months 4 to 12, with the MaternaLink build running, four or five seats.

Component	Phase A per month	Phase B per month
Domain, DNS, email security	£2	£2
Google Workspace	£10–£22	£25–£55
Website (Webflow)	£18–£25	£18–£25
Website analytics (Plausible)	£9	£9
CRM (HubSpot)	£0–£20	£20–£40
Finance (Xero, with Stripe and GoCardless)	£15–£30	£15–£30
Work management (Notion, GitHub Projects)	£0–£16	£16–£40
Contracts (PandaDoc or DocuSign)	£0–£25	£25–£40
Password manager (1Password)	£12–£16	£30–£40
Scheduling (Calendly)	£0–£10	£0–£10
AI assistants	£20–£60	£80–£200
Course platform	£0	£30–£120, from about month 6
GitHub Team	£0	£12–£20
AWS, as environments come online	£0	£150–£700
Monitoring (Sentry, Grafana, uptime checks)	£0	£20–£80
SMS gateways, sandbox then pilot	£0	£10–£150, billed on in pilots

Component	Phase A per month	Phase B per month
Tools total	£90–£250	£460–£1,560
Fractional data protection officer	£0–£500	£500–£1,500
Total including that officer	£90–£750	£960–£3,060

Do not build this yet

Item	Why not now	Build it when
A custom website or CMS	No engineers. Webflow is enough	The engineering team exists and the brand has settled
A custom course platform	Bought platforms cover years 1 and 2	Learning revenue passes £150k a year, or a buyer demands self-hosting
A customer portal	There are no customers to serve	Five or more paying organisations
A data warehouse	The volumes are tiny. Metabase and a spreadsheet are enough	More than five organisations, or research data needs it
Native mobile apps	The web app meets the need. App stores add clinical safety re-testing on every release	Evaluation shows the web app cannot do it
NHS interfaces and FHIR	There is no NHS customer, and onboarding is heavy	A trust asks for it, in version 2
Hosting inside Nigeria	No contract requires it	A signed contract has a data residency clause
Two live regions at once	Cost and complexity	A contract demands better than 99.9% availability
AI on patient data	No DPIA, no in-region endpoint, no clinical review process	Version 1.1 at the earliest, under the written policy
ISO 27001	NHS buyers ask for Cyber Essentials Plus and the toolkit first	A buyer requires it, or year 3
Kubernetes, or splitting into microservices	Managed containers and one well-organised application are faster and safer	Many services with someone to run them, or a team past ten engineers
A separate sign-in service for women	Phone code plus PIN is enough	Merging it with staff sign-in is clearly worth it
Blockchain, tokens, own-brand wearables	No user needs them and they damage credibility	Never, unless evidence appears

Rules that do not bend

1. Patient data lives only in the MaternaLink platform, in London, behind row-level security, audit and encryption. Business tools never receive it.
2. Every outside service is a sub-processor: it needs an agreement, a register entry and a DPIA update.
3. Every integration sits behind an interface, so a customer's choice of SMS provider or hospital system is configuration, not code.
4. Customer exports use documented formats, so nobody is locked in.
5. Monitoring, backups and infrastructure-as-code start with the first environment. They are the cheapest assurance evidence you will ever buy.

Words explained

Word	What it means
CMS	The tool that lets a non-developer edit website pages. A headless CMS stores the content separately from the site that displays it
CRM	The tool that tracks who you have spoken to and what you sold
Row-level security	The database itself refuses to show one customer's rows to another
Sub-processor	Any outside company that touches your data on your behalf
Data residency	The country the data physically sits in
Infrastructure-as-code	The servers are described in files, so they can be rebuilt exactly
FHIR and DHIS2	The standard NHS systems use to exchange health records, and the reporting system most African health ministries run
ICO	The UK data protection regulator. You register with it on incorporation
ESTIMATE / TO VALIDATE	Our best guess, not a fact / nobody has checked this yet